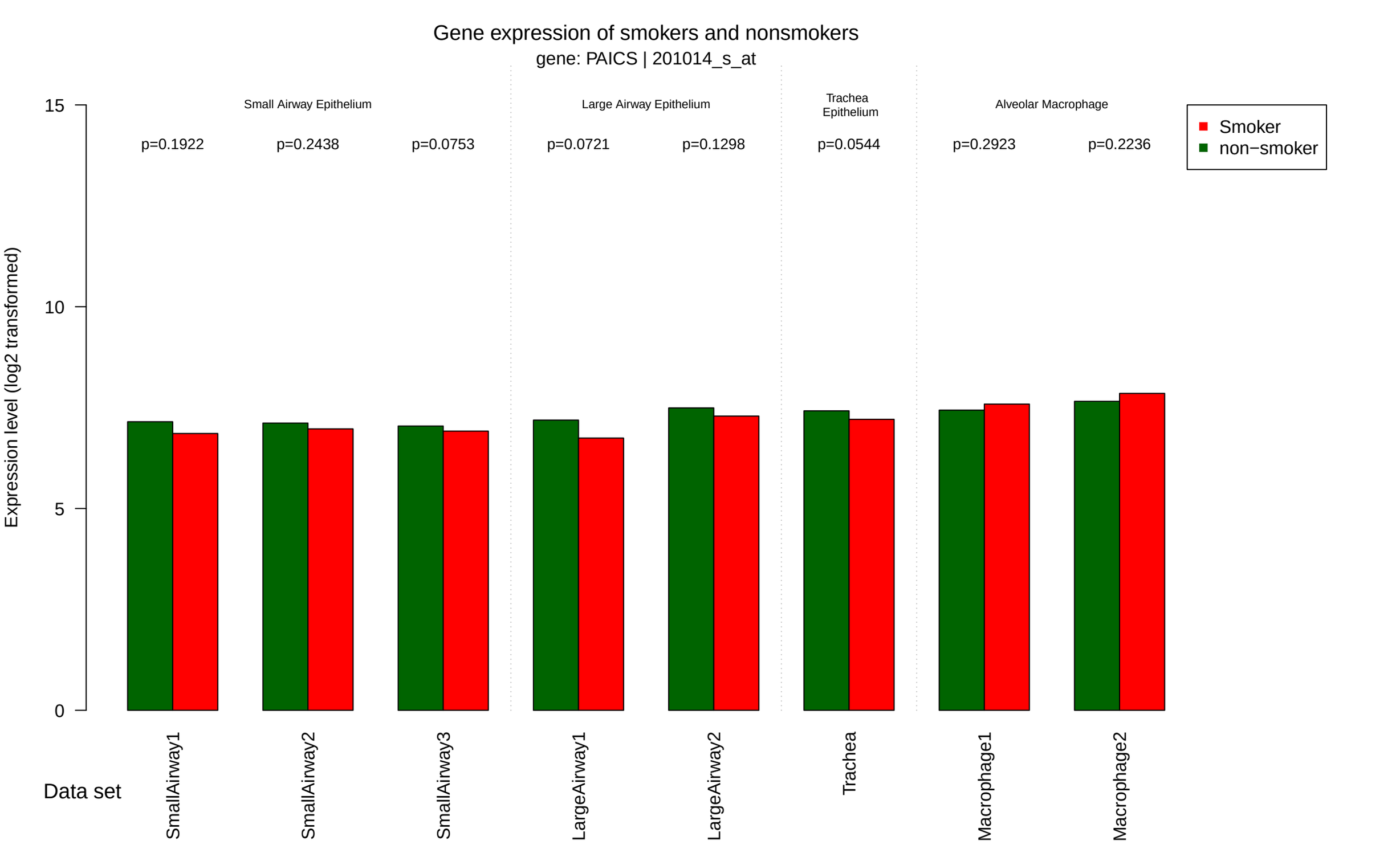
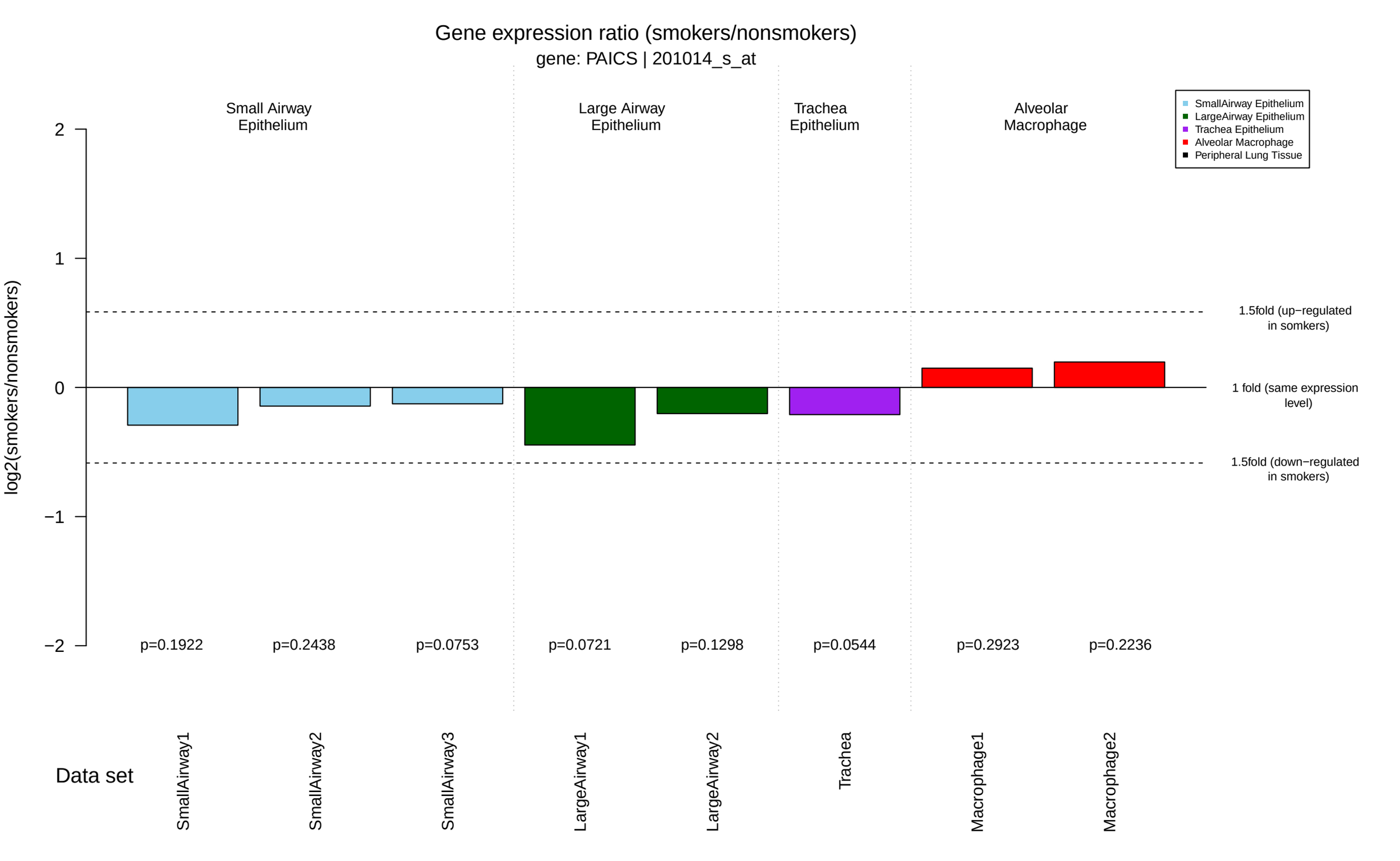
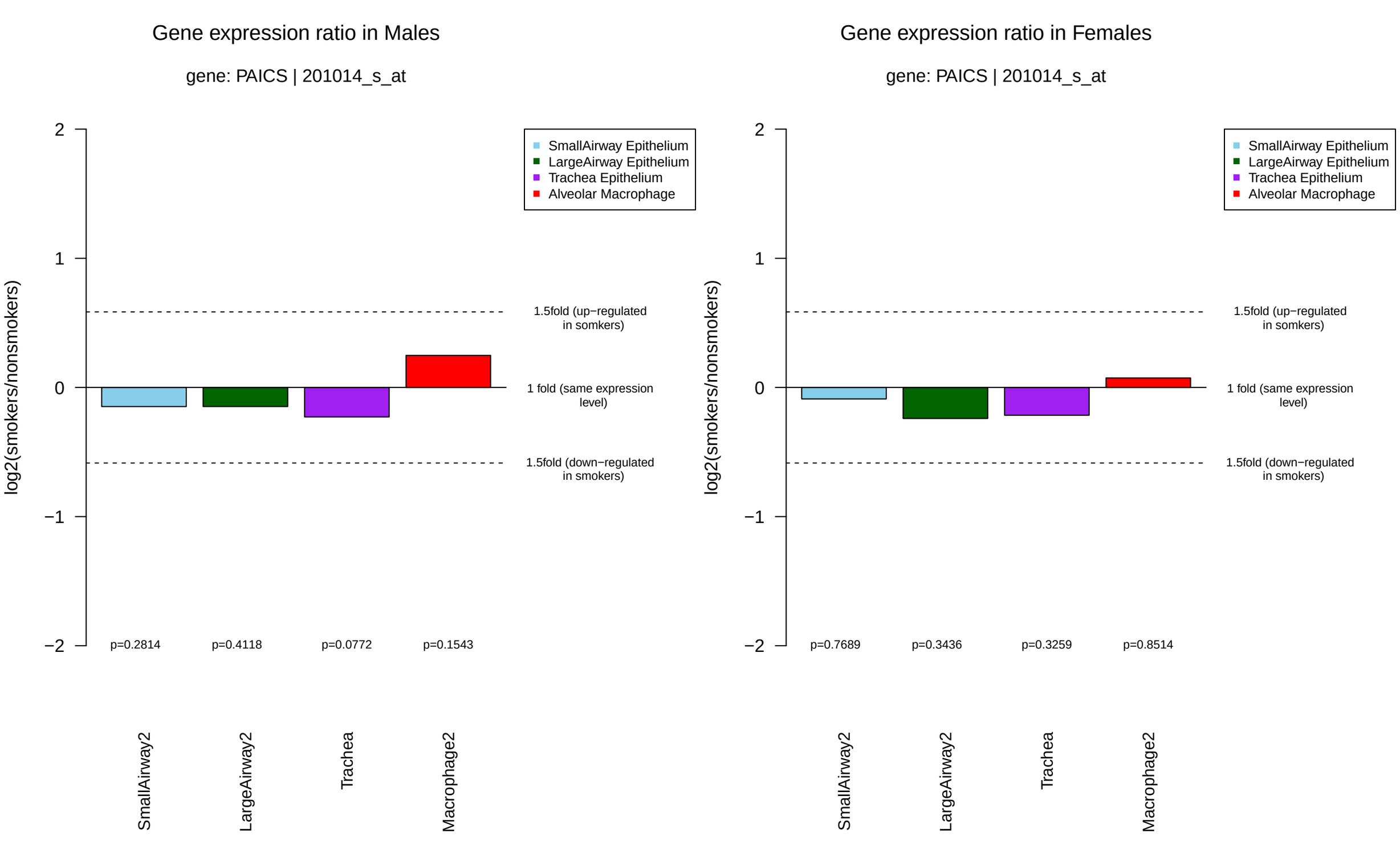
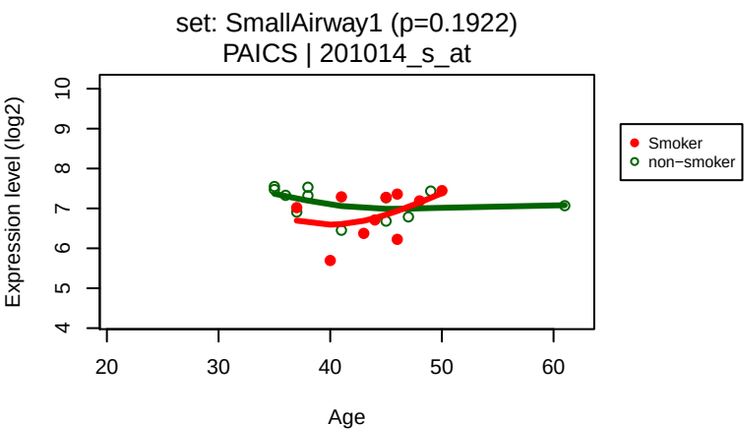
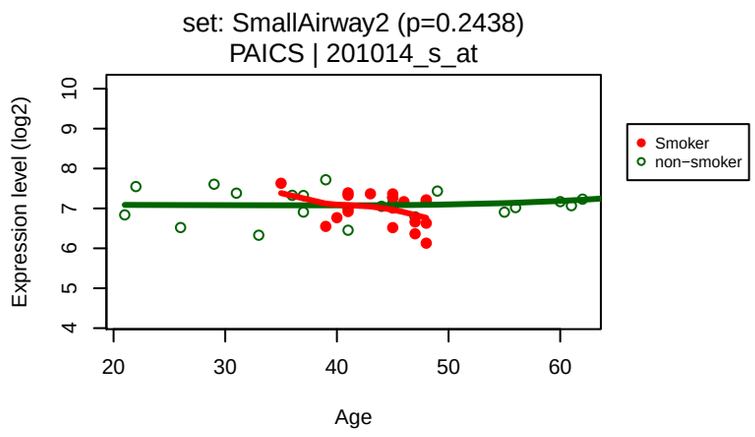
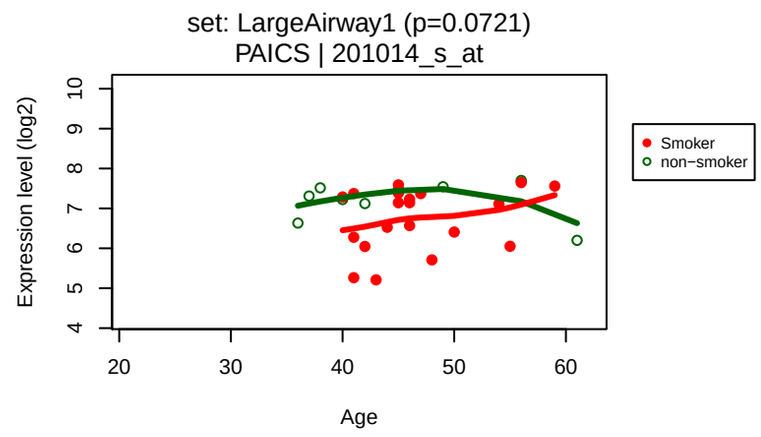

Gene expression of smokers and nonsmokers
gene: PAICS | 201014_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1922
p=0.2438
p=0.0753
p=0.0721
p=0.1298
p=0.0544
p=0.2923
p=0.2236
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

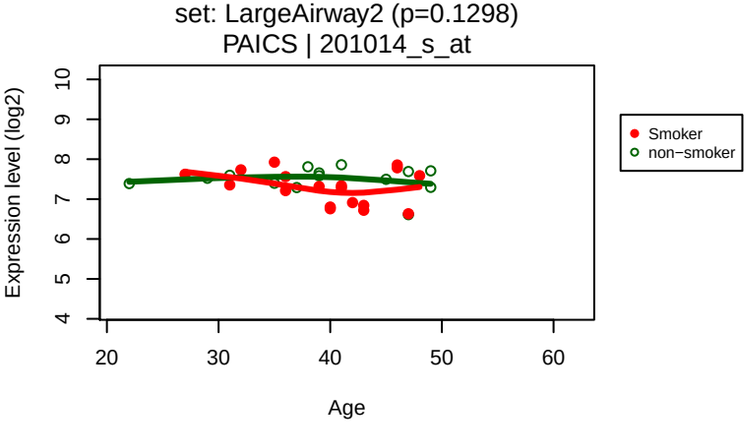

Gene expression ratio (smokers/nonsmokers)
gene: PAICS | 201014_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1922
p=0.2438
p=0.0753
p=0.0721
p=0.1298
p=0.0544
p=0.2923
p=0.2236
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

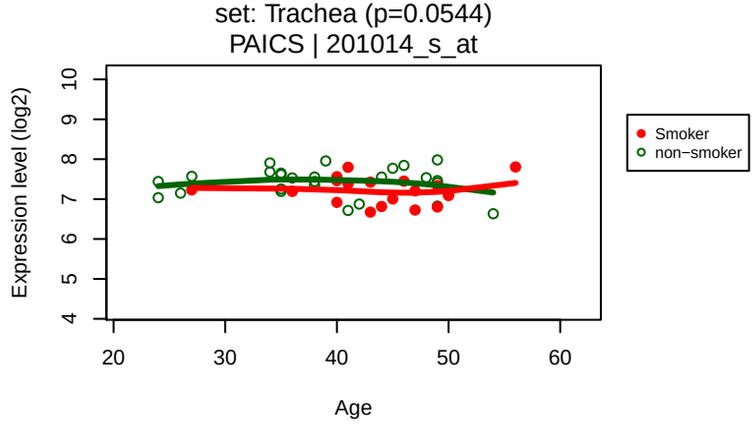

Gene expression ratio in Males
gene: PAICS | 201014_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2814
p=0.4118
p=0.0772
p=0.1543
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAICS | 201014_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7689
p=0.3436
p=0.3259
p=0.8514
SmallAirway2
LargeAirway2
Trachea
Macrophage2

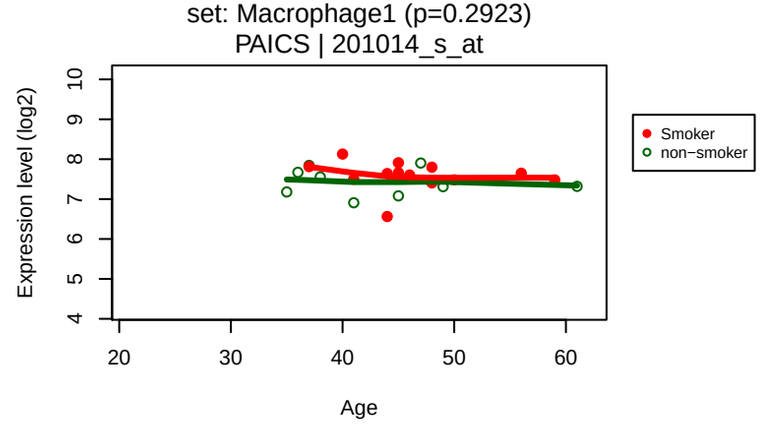

set: SmallAirway1 (p=0.1922)
PAICS | 201014_s_at
Expression level (log2)
Age
Smoker
non−smoker

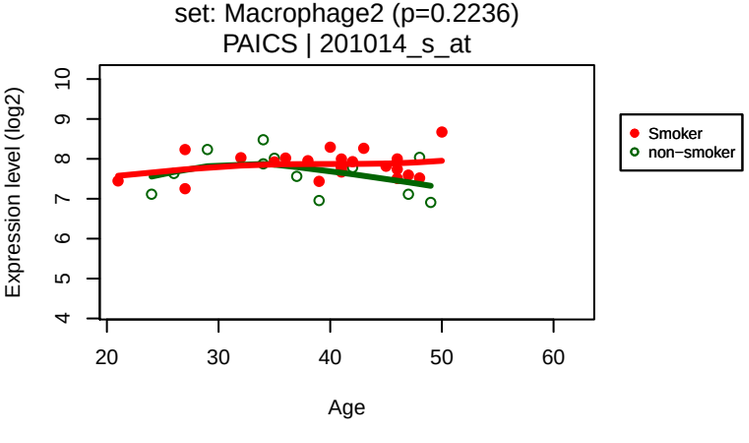

set: SmallAirway2 (p=0.2438)
PAICS | 201014_s_at
Expression level (log2)
Age
Smoker
non−smoker

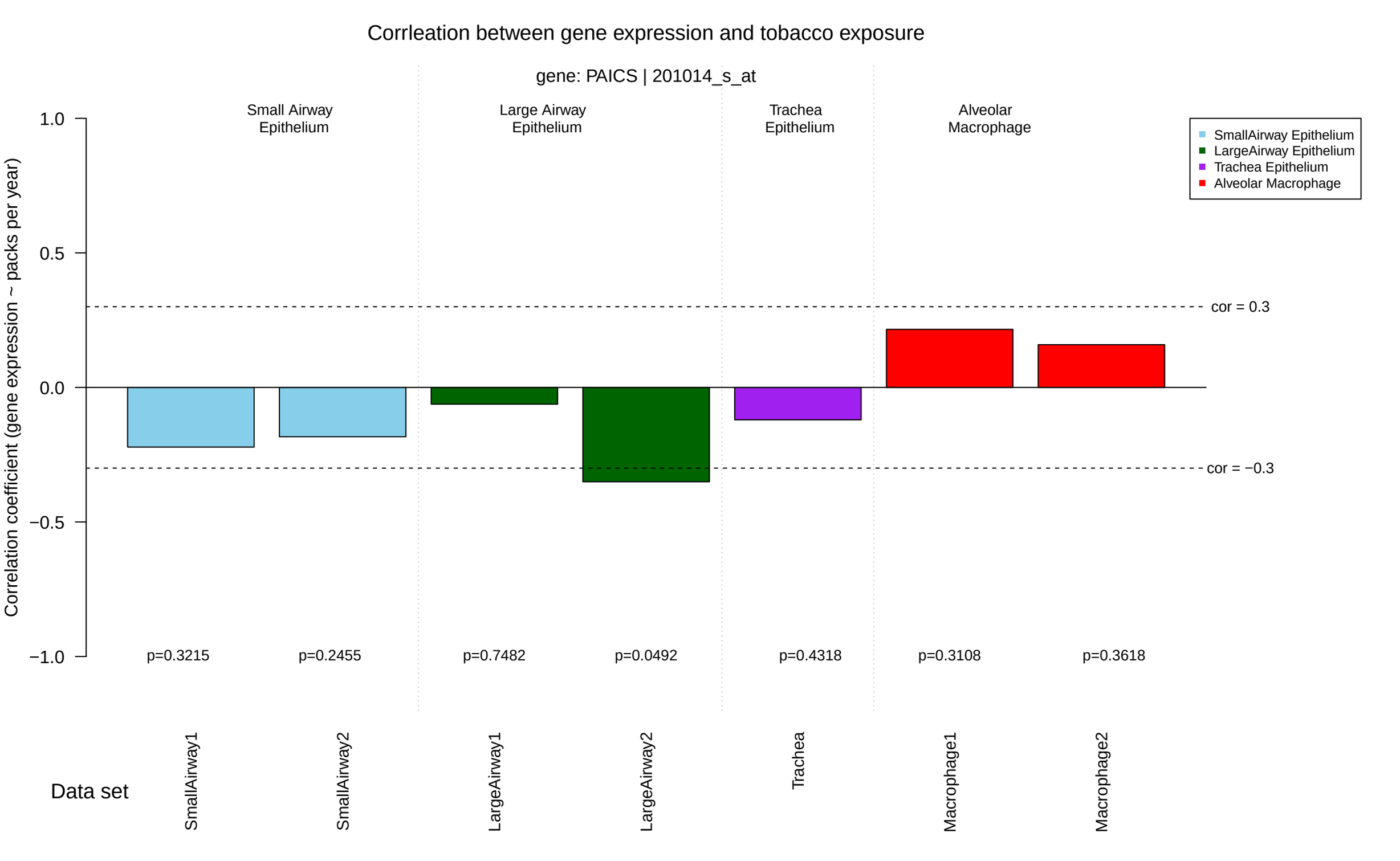

set: LargeAirway1 (p=0.0721)
PAICS | 201014_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1298)
PAICS | 201014_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0544)
PAICS | 201014_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2923)
PAICS | 201014_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2236)
PAICS | 201014_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAICS | 201014_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3215
p=0.2455
p=0.7482
p=0.0492
p=0.4318
p=0.3108
p=0.3618
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2